

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

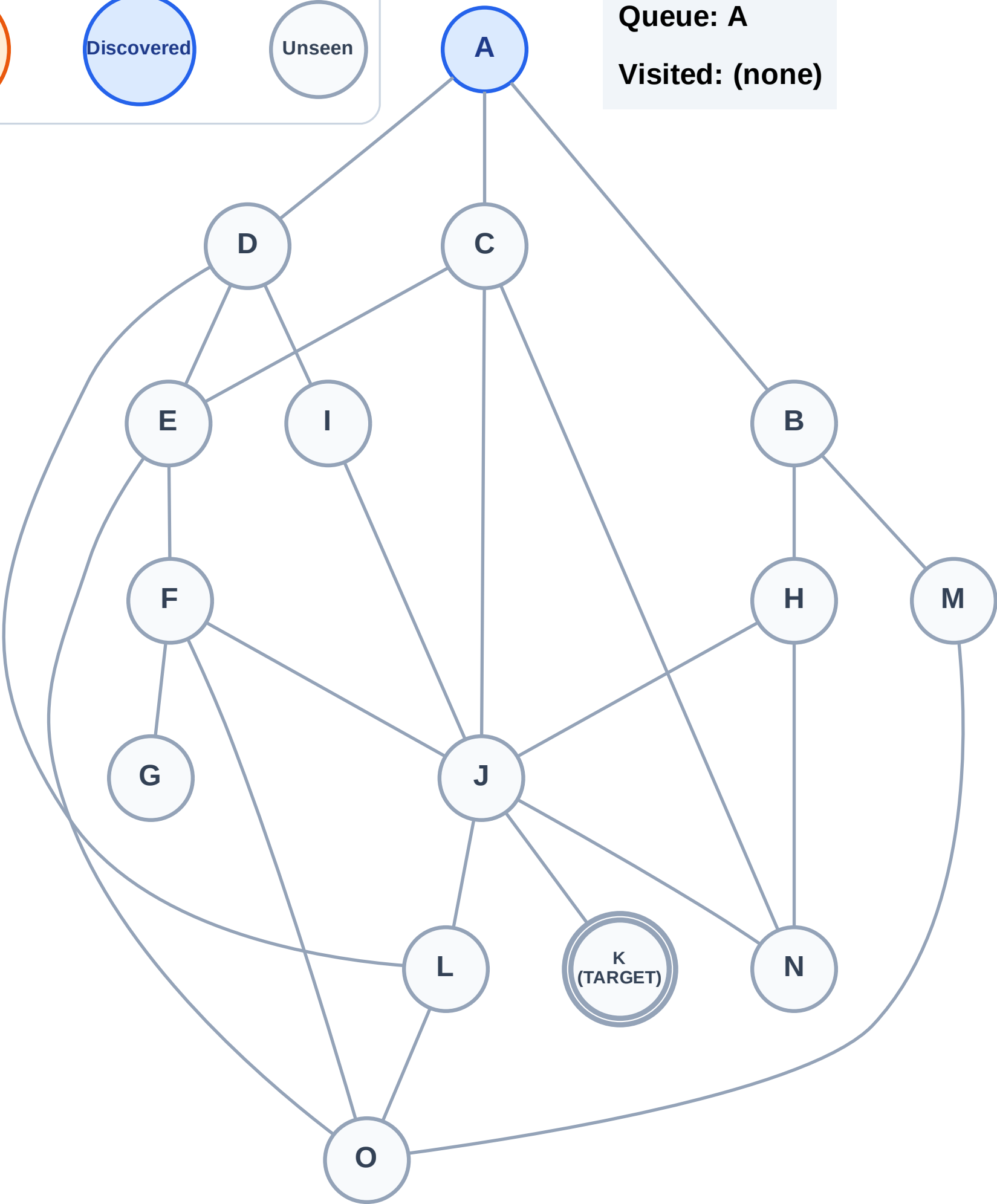
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

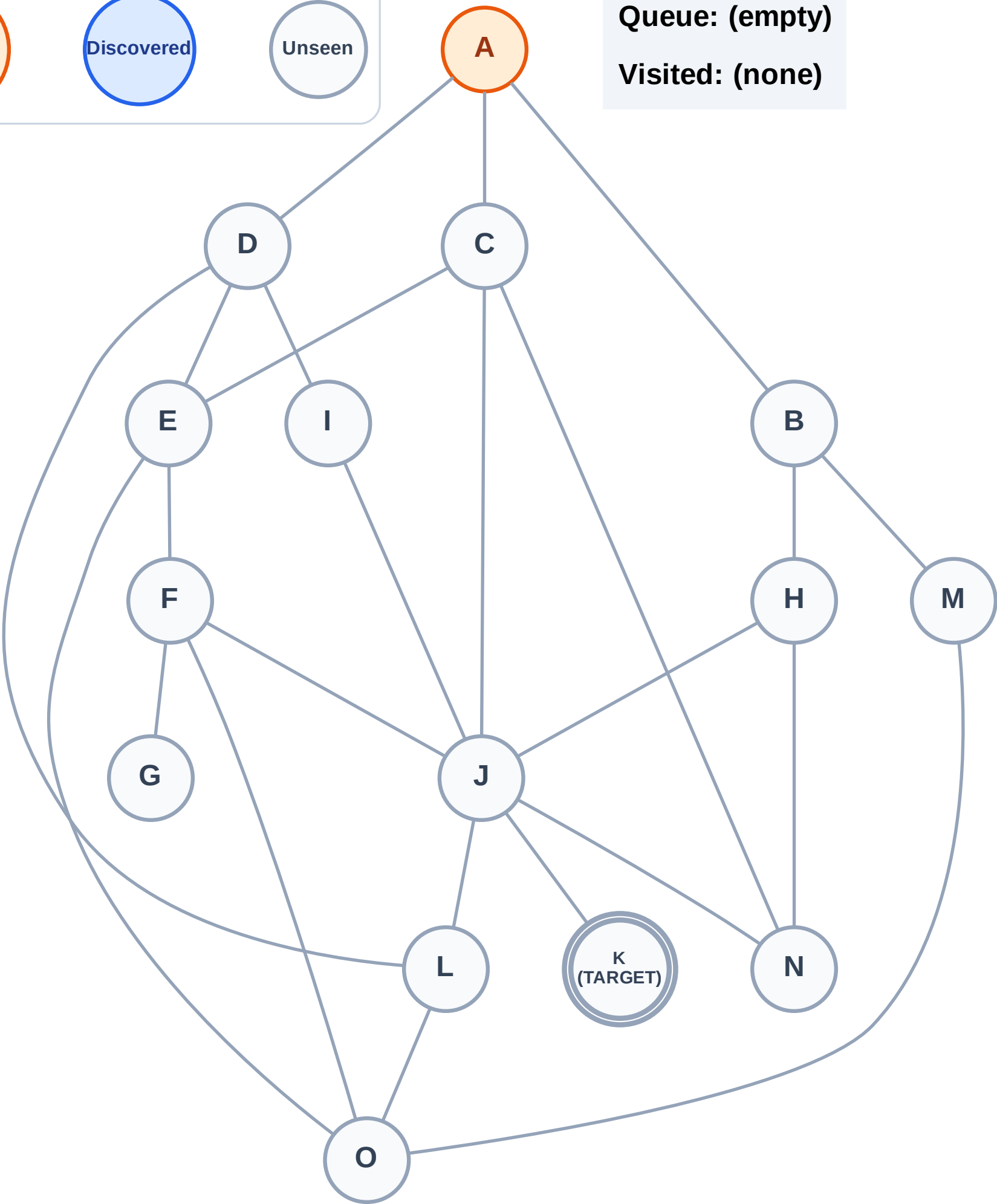
Complete

Processing

Discovered

Unseen

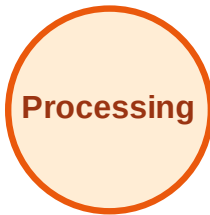
Queue: (empty)
Visited: (none)



BFS Traversal

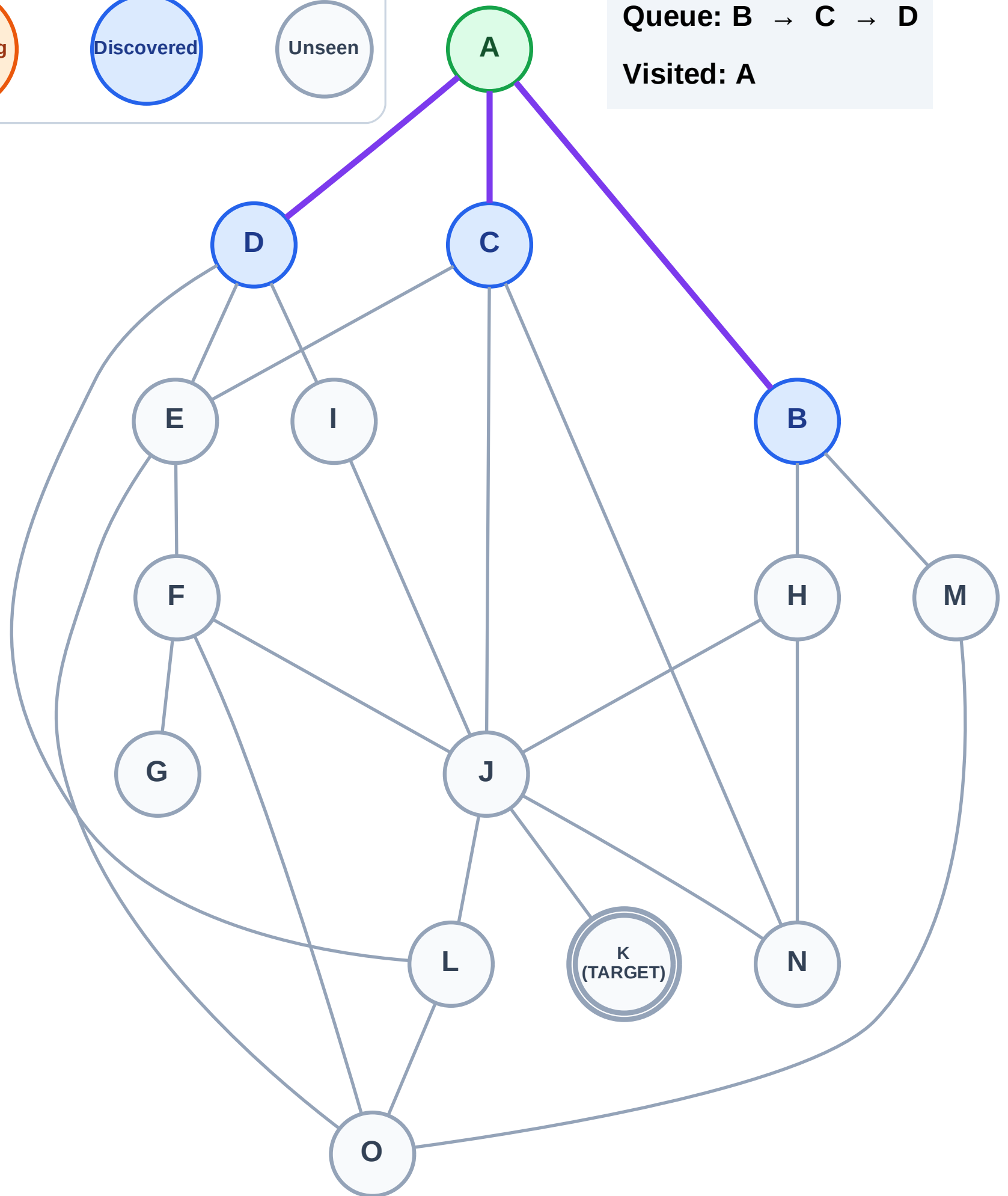
Step 3: Discover B, C, D from A

Legend



Queue: B → C → D

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

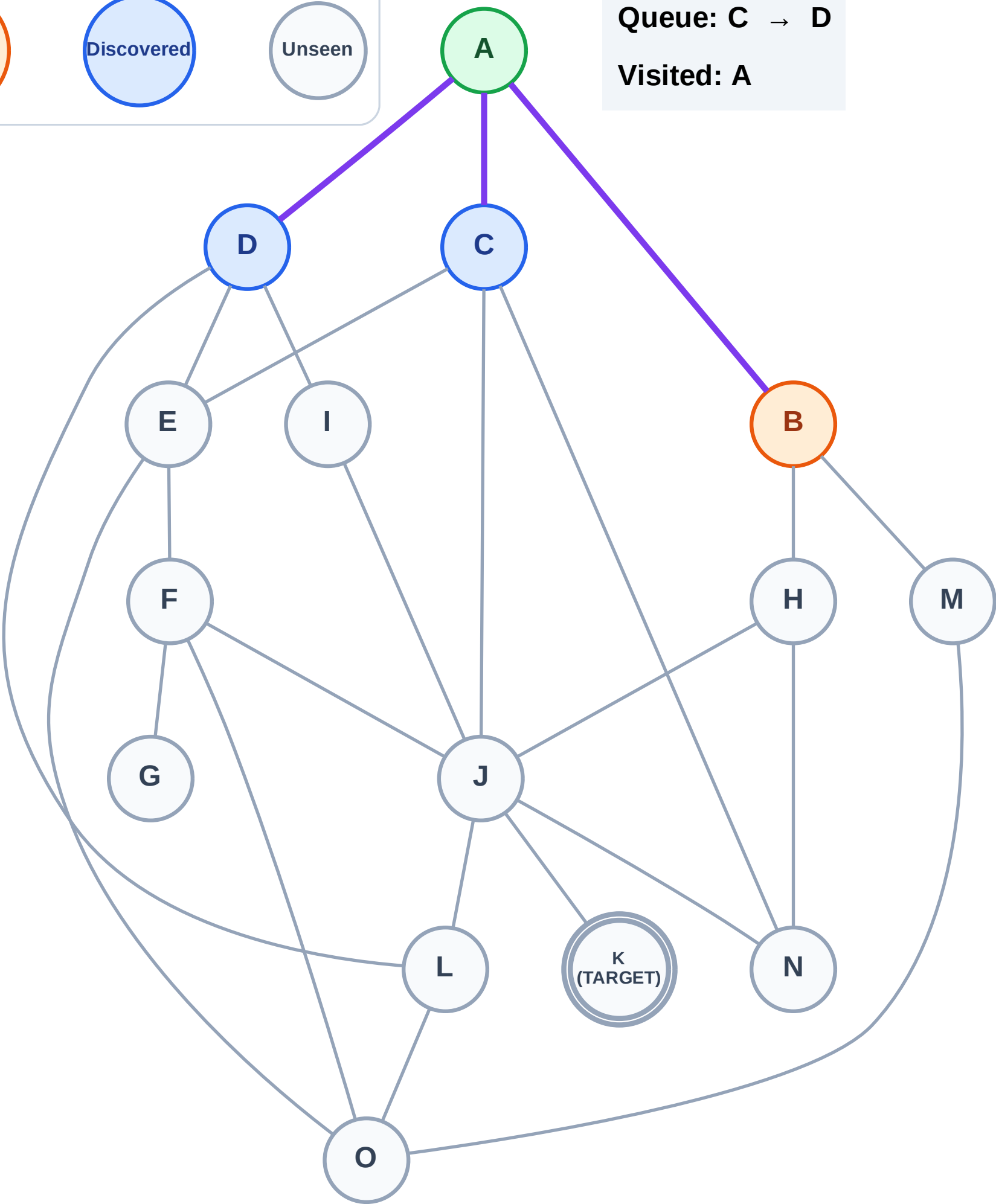
Processing

Discovered

Unseen

Queue: C → D

Visited: A



BFS Traversal

Step 5: Discover H, M from B

Legend

Complete

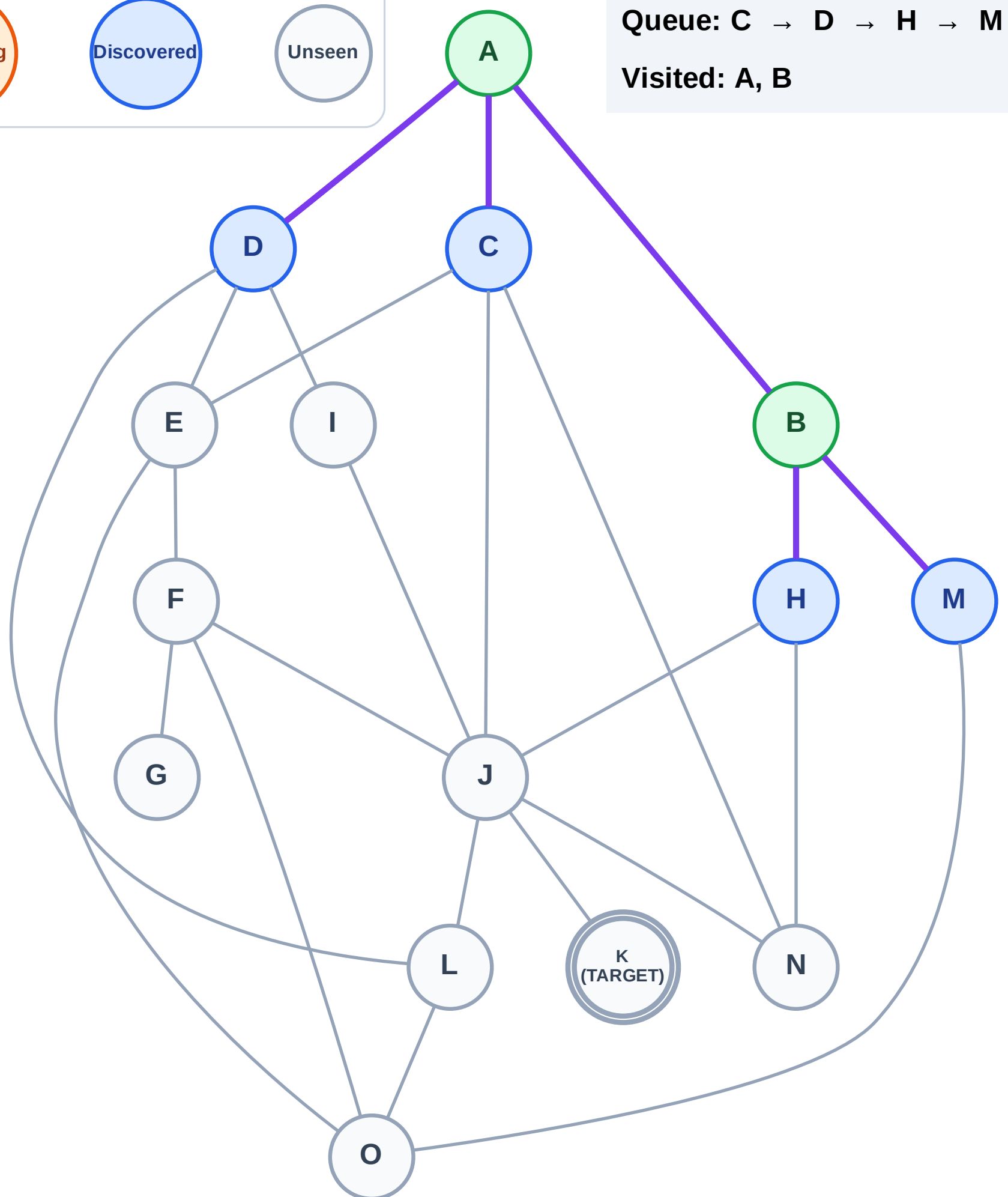
Processing

Discovered

Unseen

Queue: C → D → H → M

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

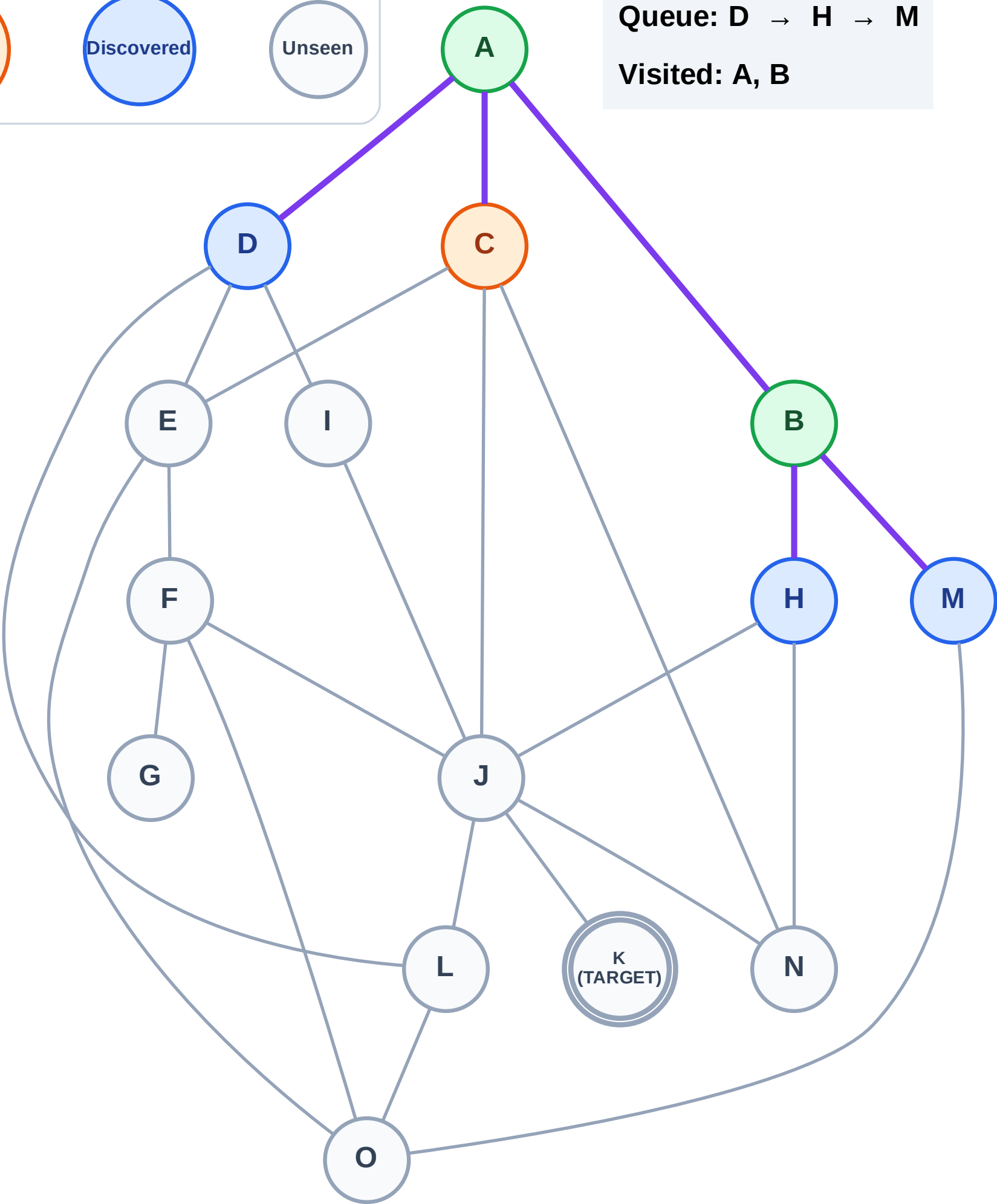
Processing

Discovered

Unseen

Queue: D → H → M

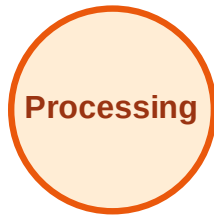
Visited: A, B



BFS Traversal

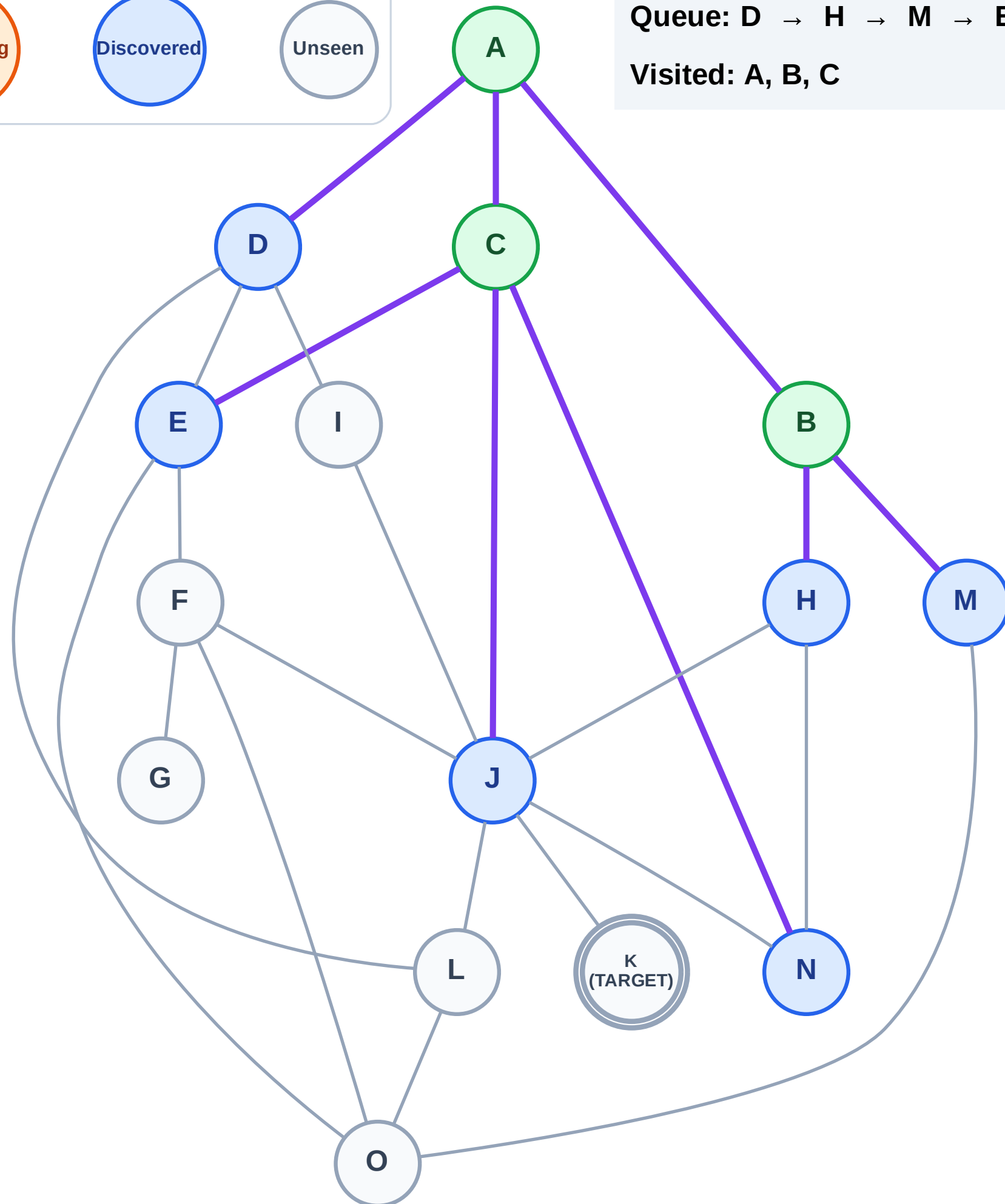
Step 7: Discover E, J, N from C

Legend



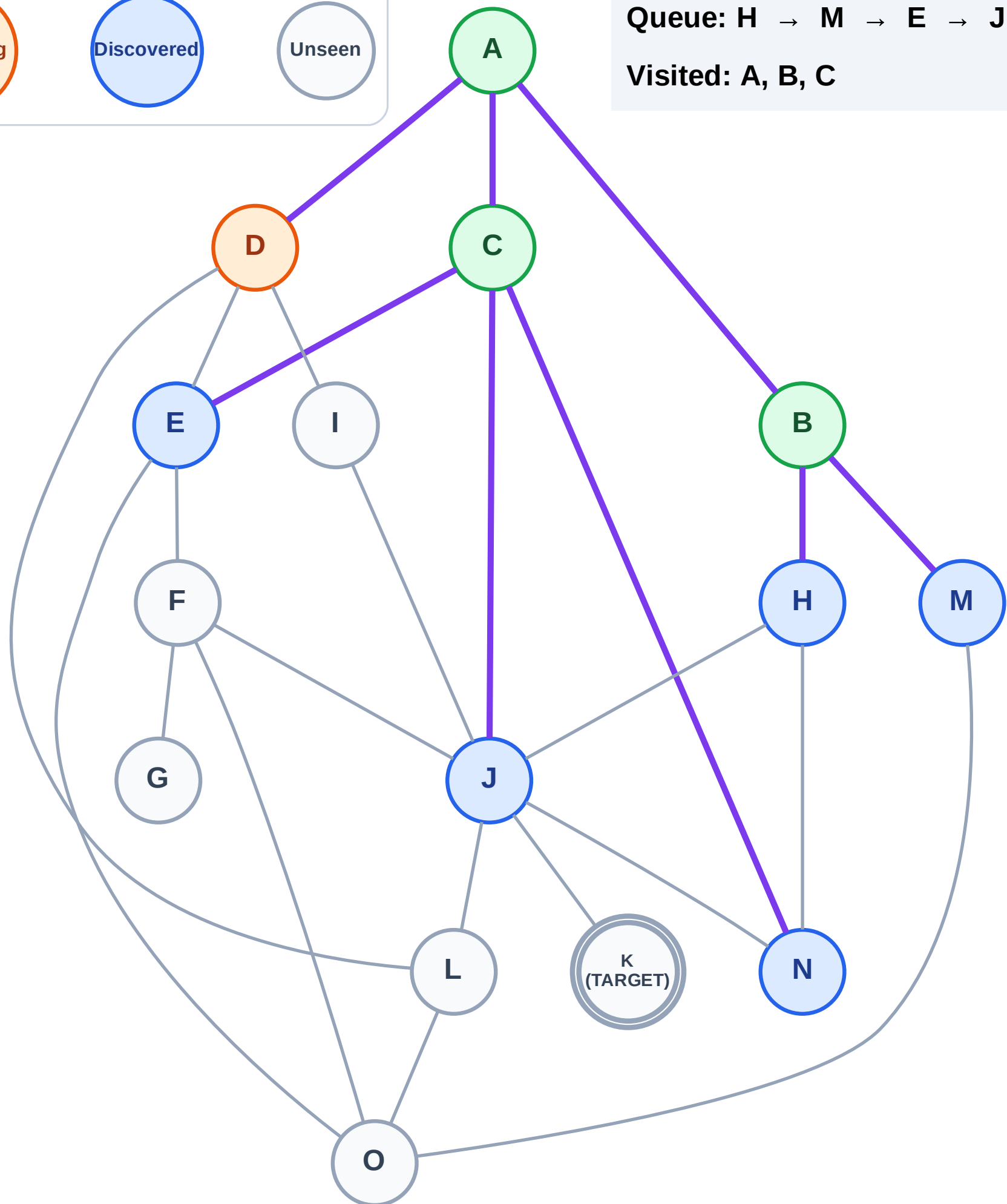
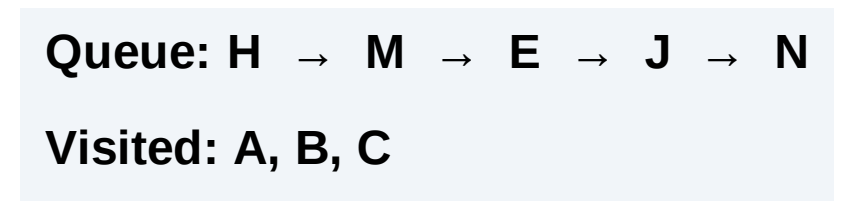
Queue: D → H → M → E → J → N

Visited: A, B, C



Step 8: Process node D

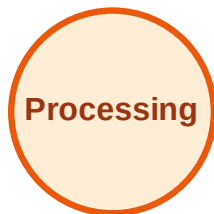
Page 10 of 10



BFS Traversal

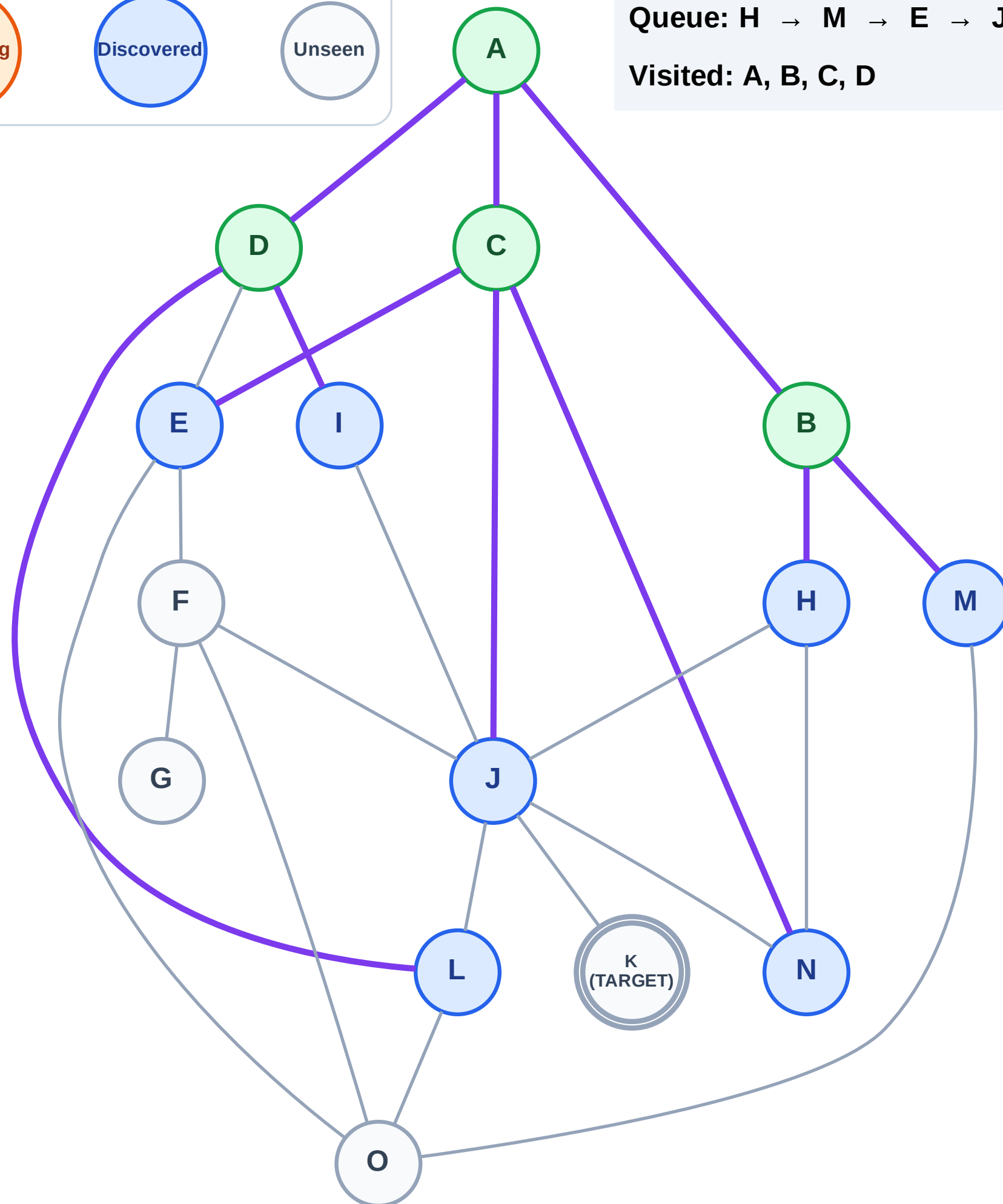
Step 9: Discover I, L from D

Legend



Queue: H → M → E → J → N → I → L

Visited: A, B, C, D



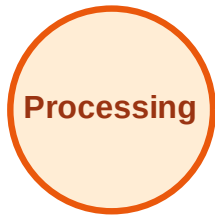
BFS Traversal

Step 10: Process node H

Legend



Complete



Processing



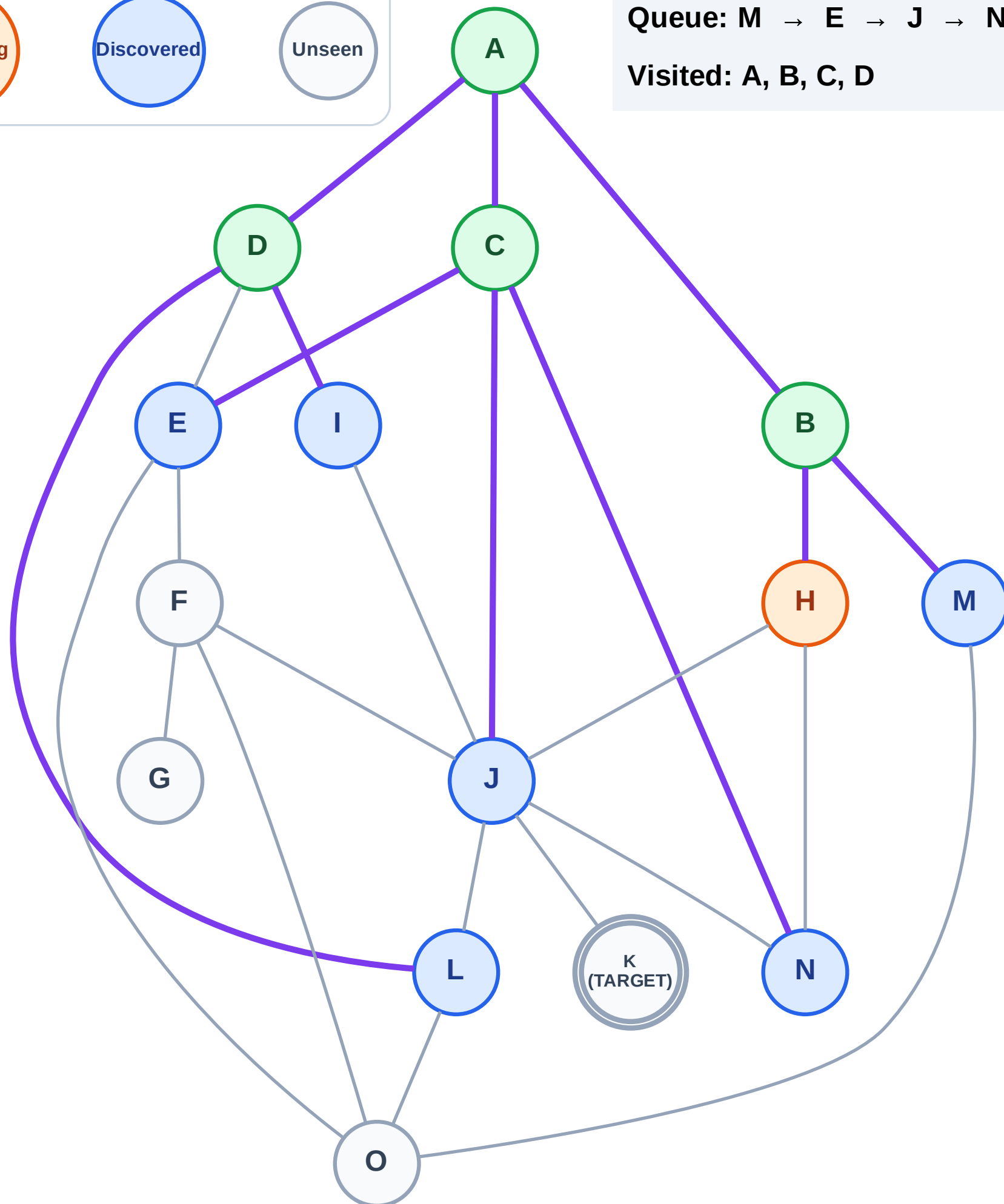
Discovered



Unseen

Queue: M → E → J → N → I → L

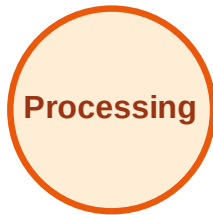
Visited: A, B, C, D



BFS Traversal

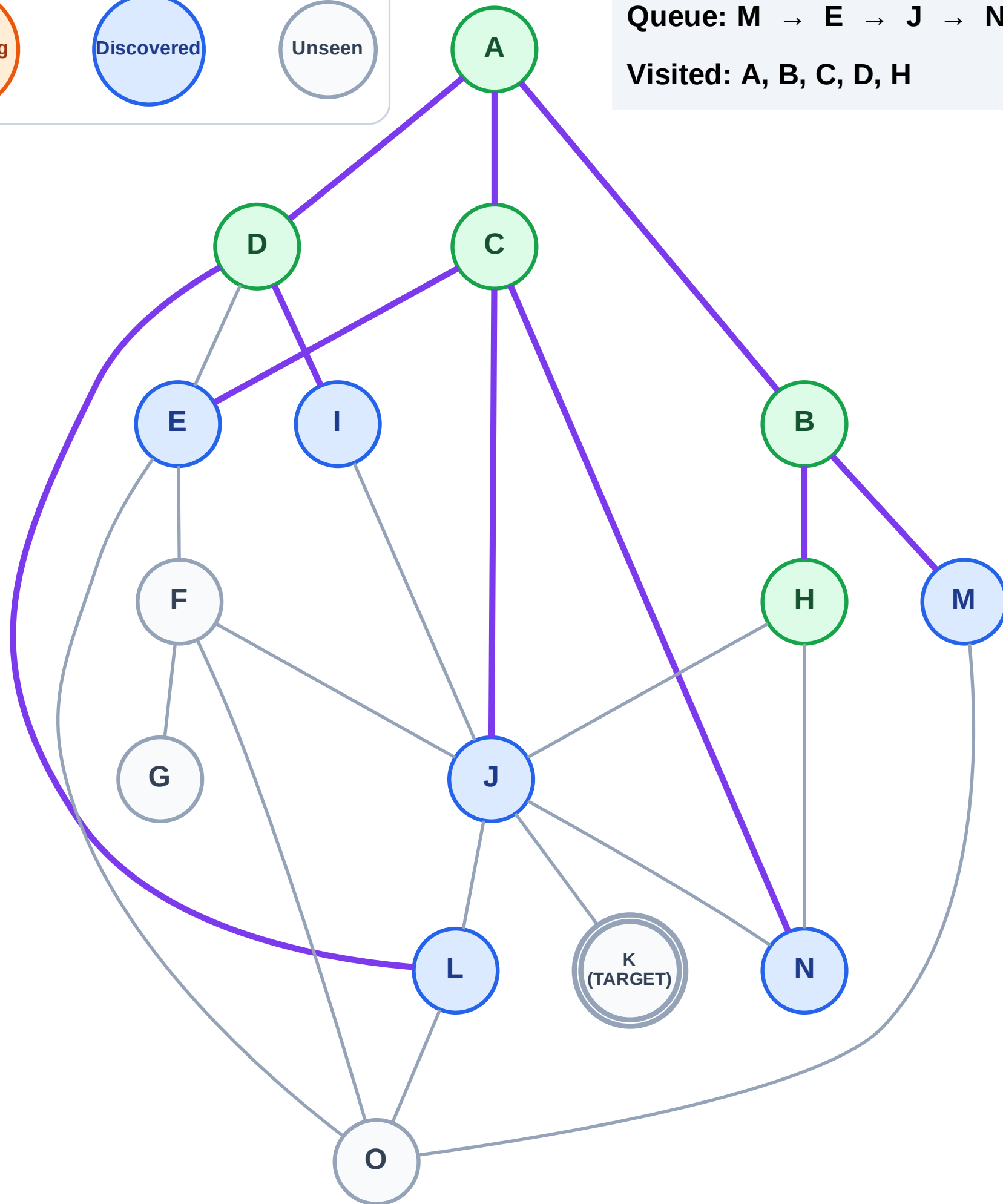
Step 11: No new neighbors from H

Legend



Queue: M → E → J → N → I → L

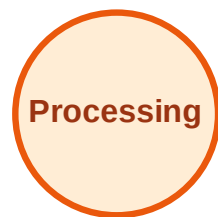
Visited: A, B, C, D, H



BFS Traversal

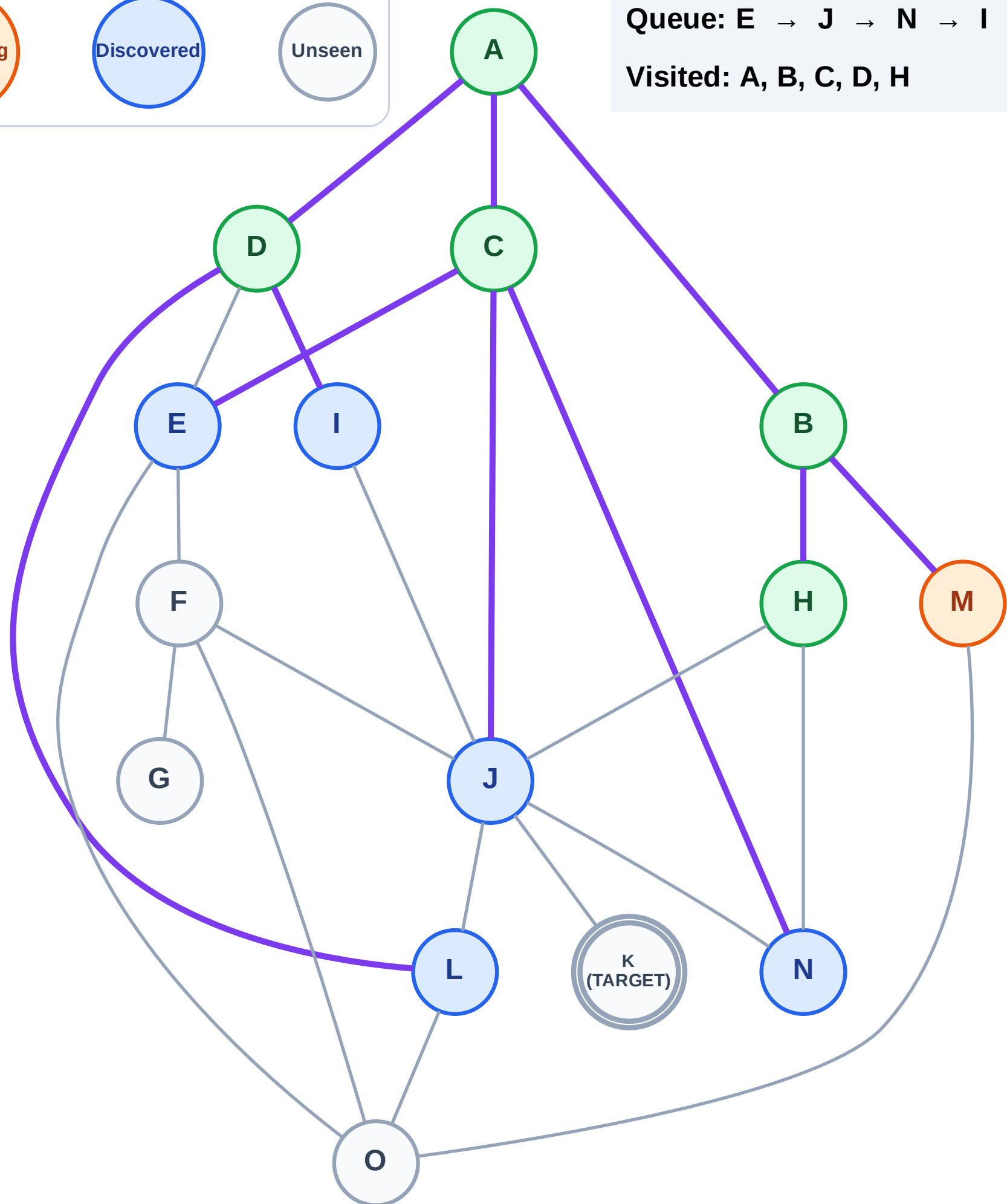
Step 12: Process node M

Legend



Queue: E → J → N → I → L

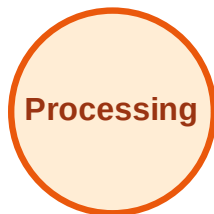
Visited: A, B, C, D, H



BFS Traversal

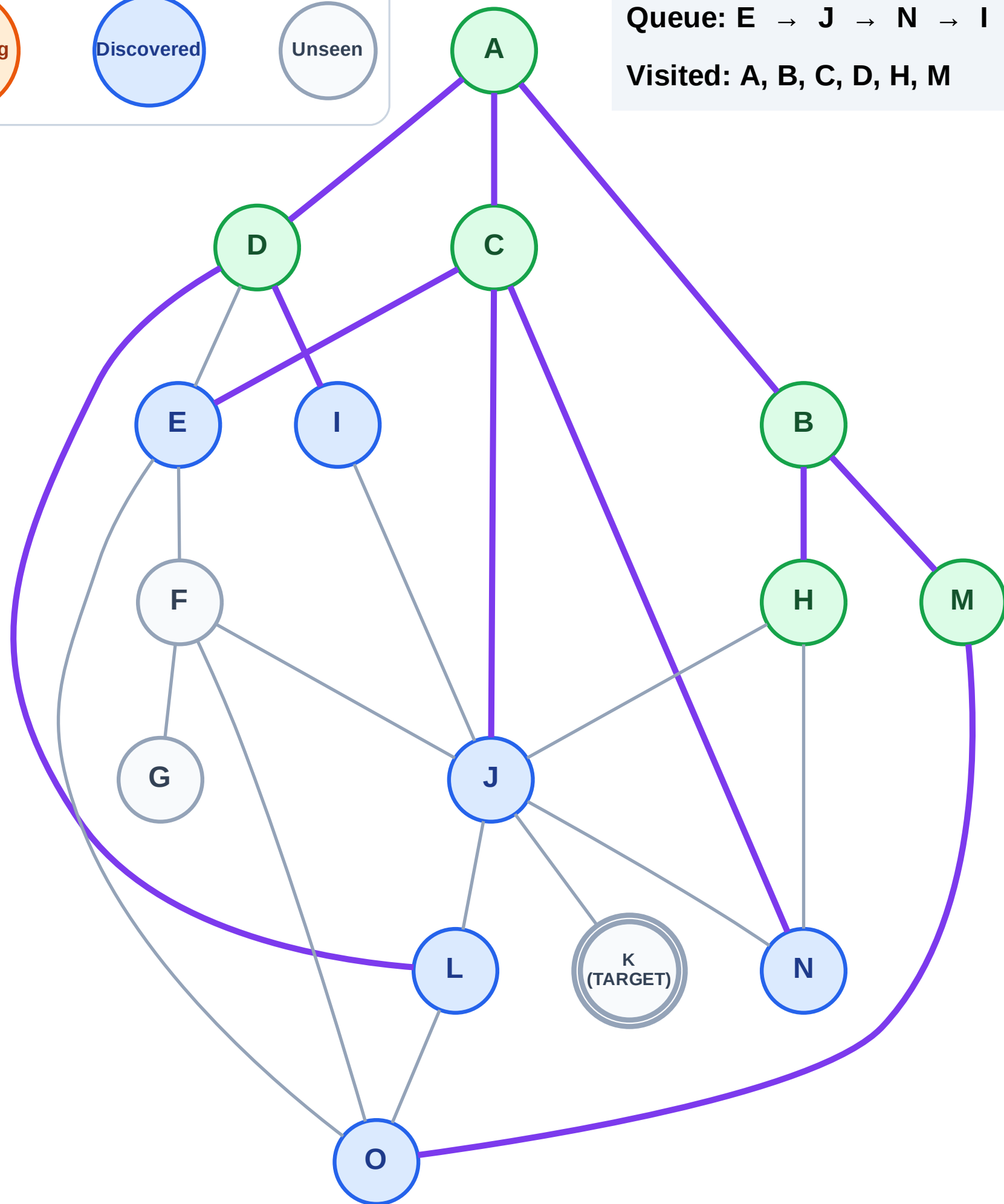
Step 13: Discover O from M

Legend



Queue: E → J → N → I → L → O

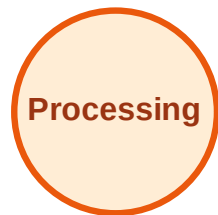
Visited: A, B, C, D, H, M



BFS Traversal

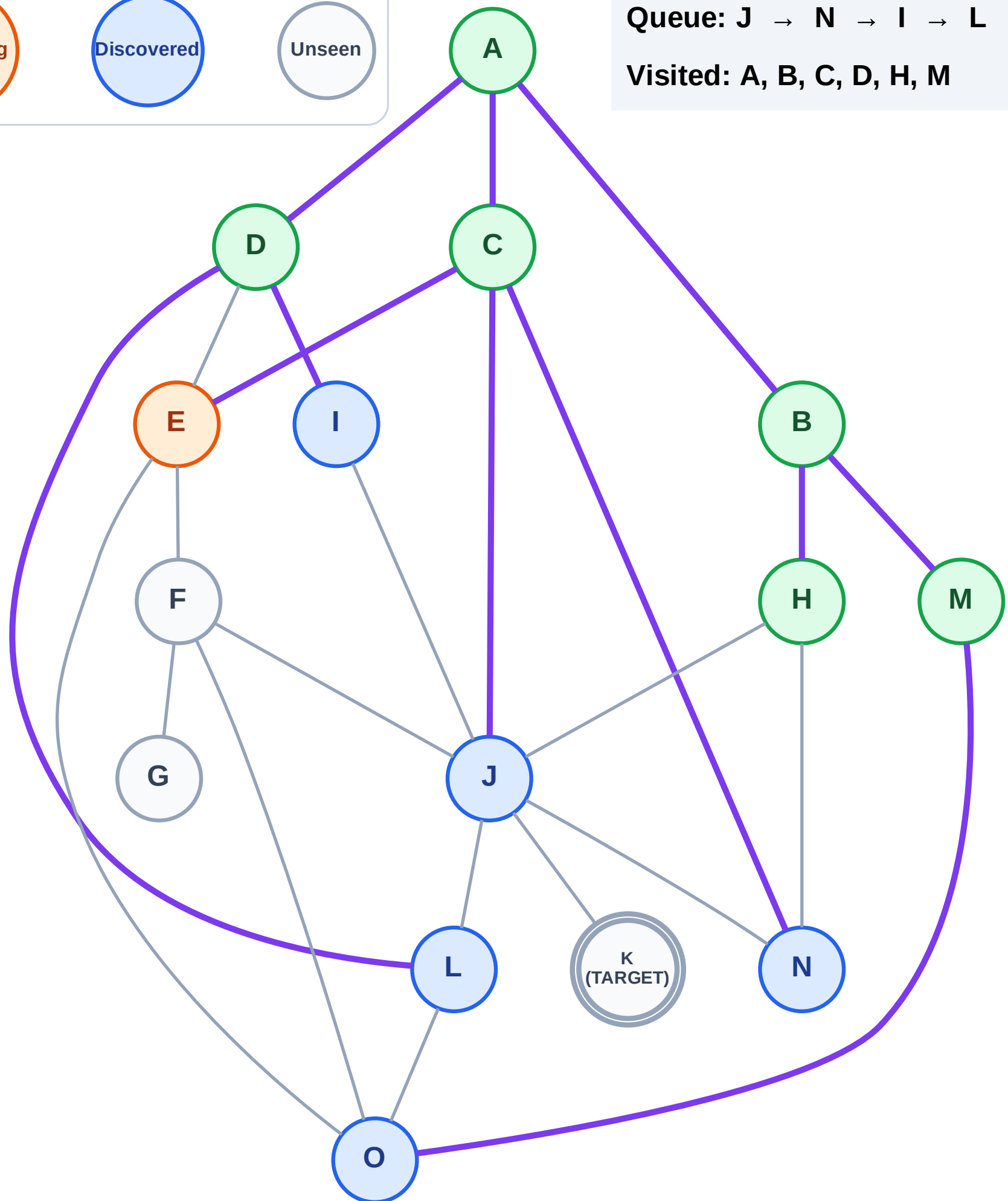
Step 14: Process node E

Legend



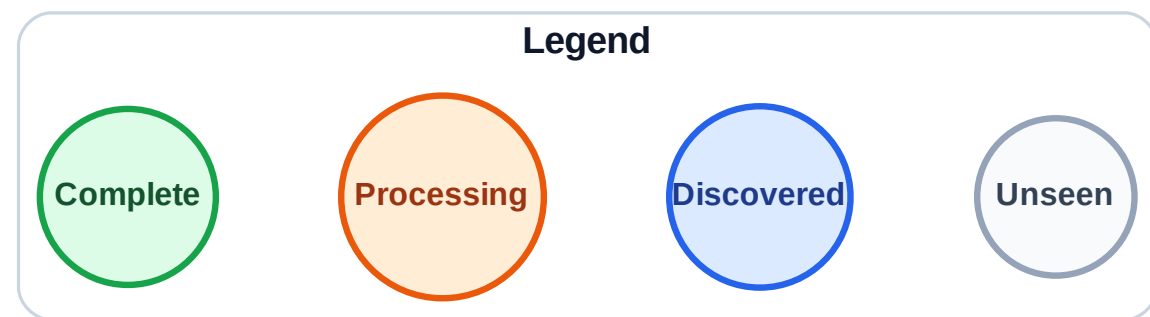
Queue: J → N → I → L → O

Visited: A, B, C, D, H, M



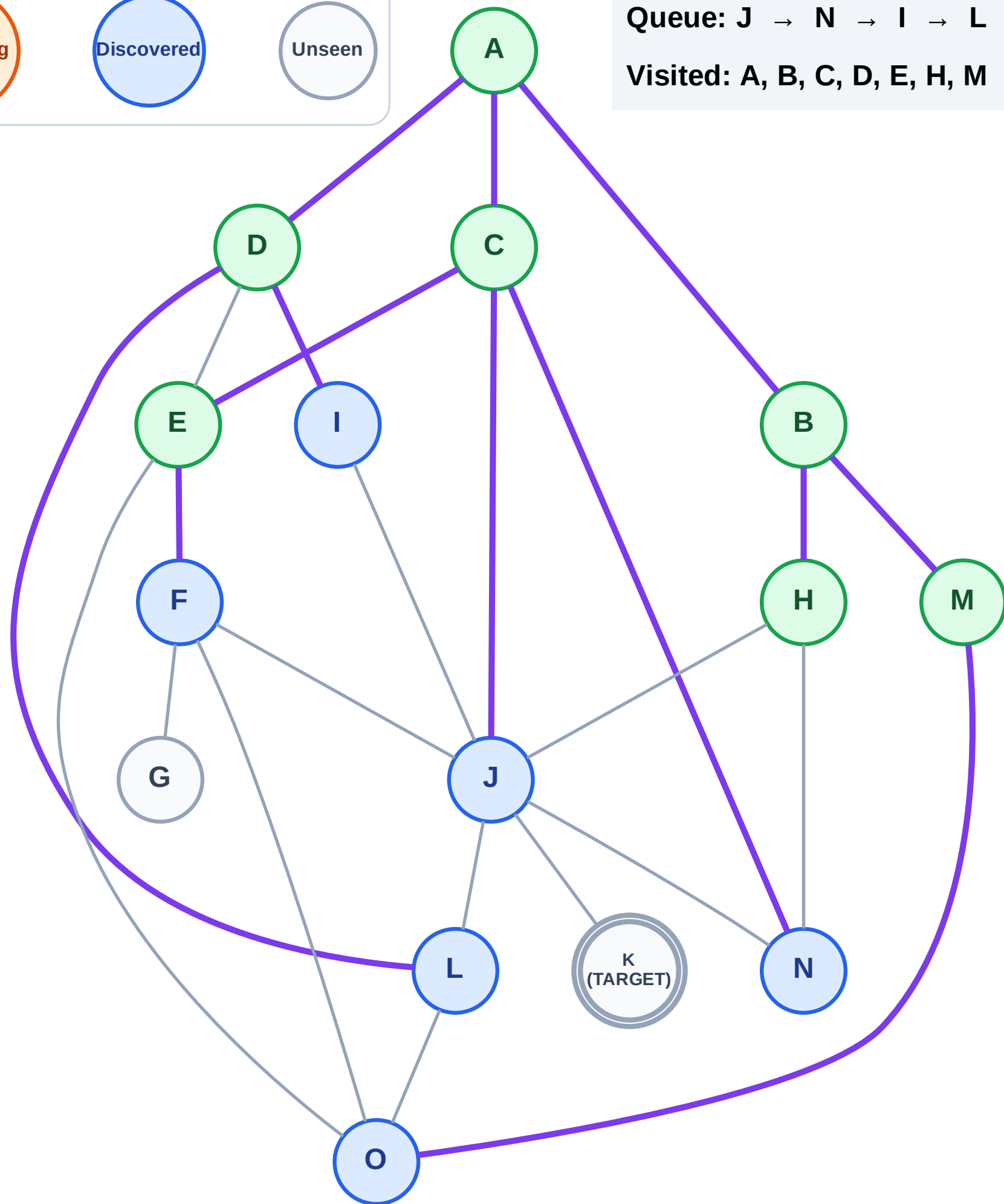
BFS Traversal

Step 15: Discover F from E



Queue: J → N → I → L → O → F

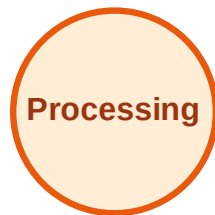
Visited: A, B, C, D, E, H, M



BFS Traversal

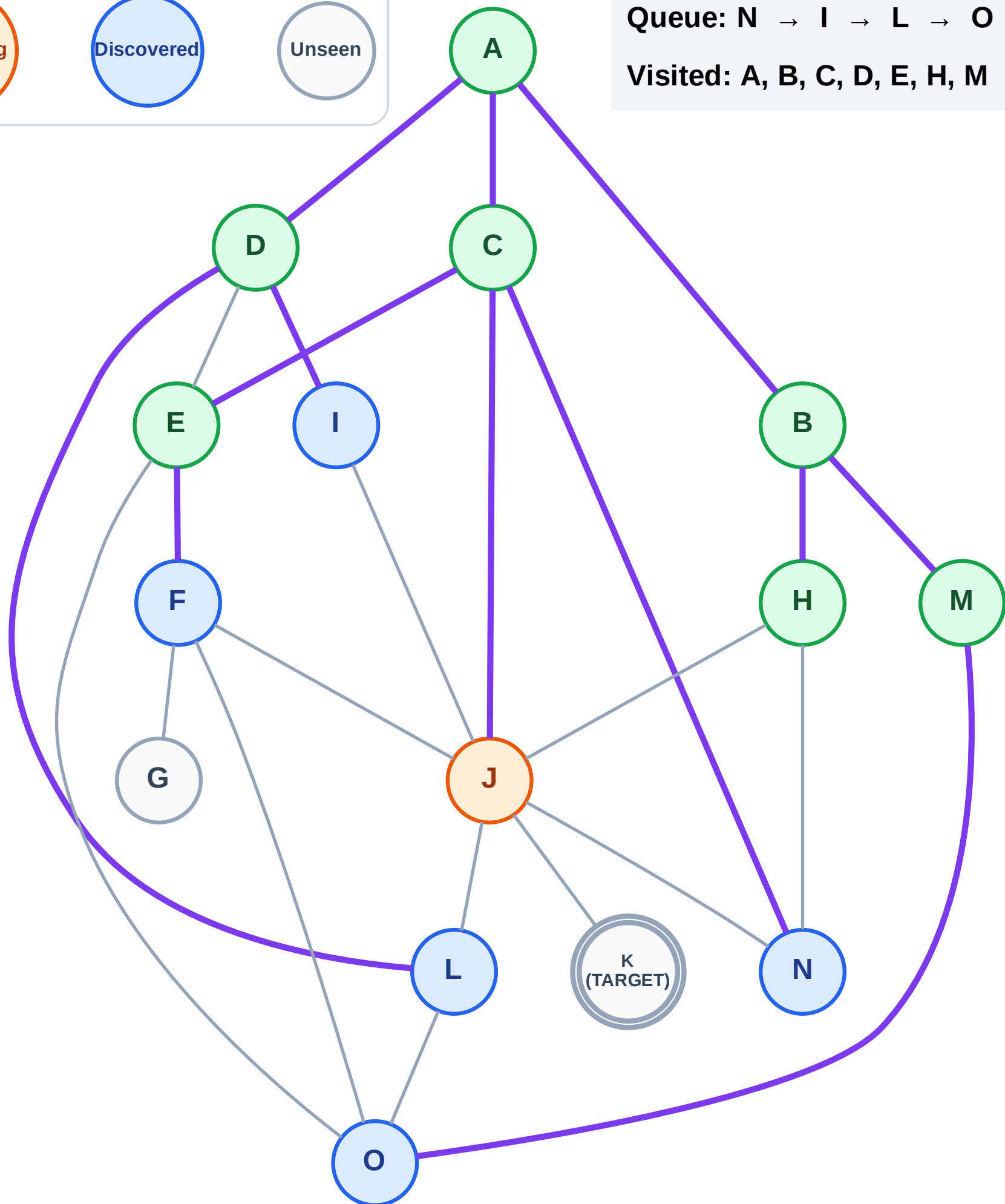
Step 16: Process node J

Legend



Queue: N → I → L → O → F

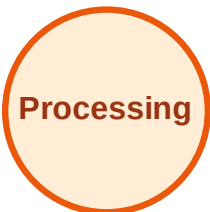
Visited: A, B, C, D, E, H, M



BFS Traversal

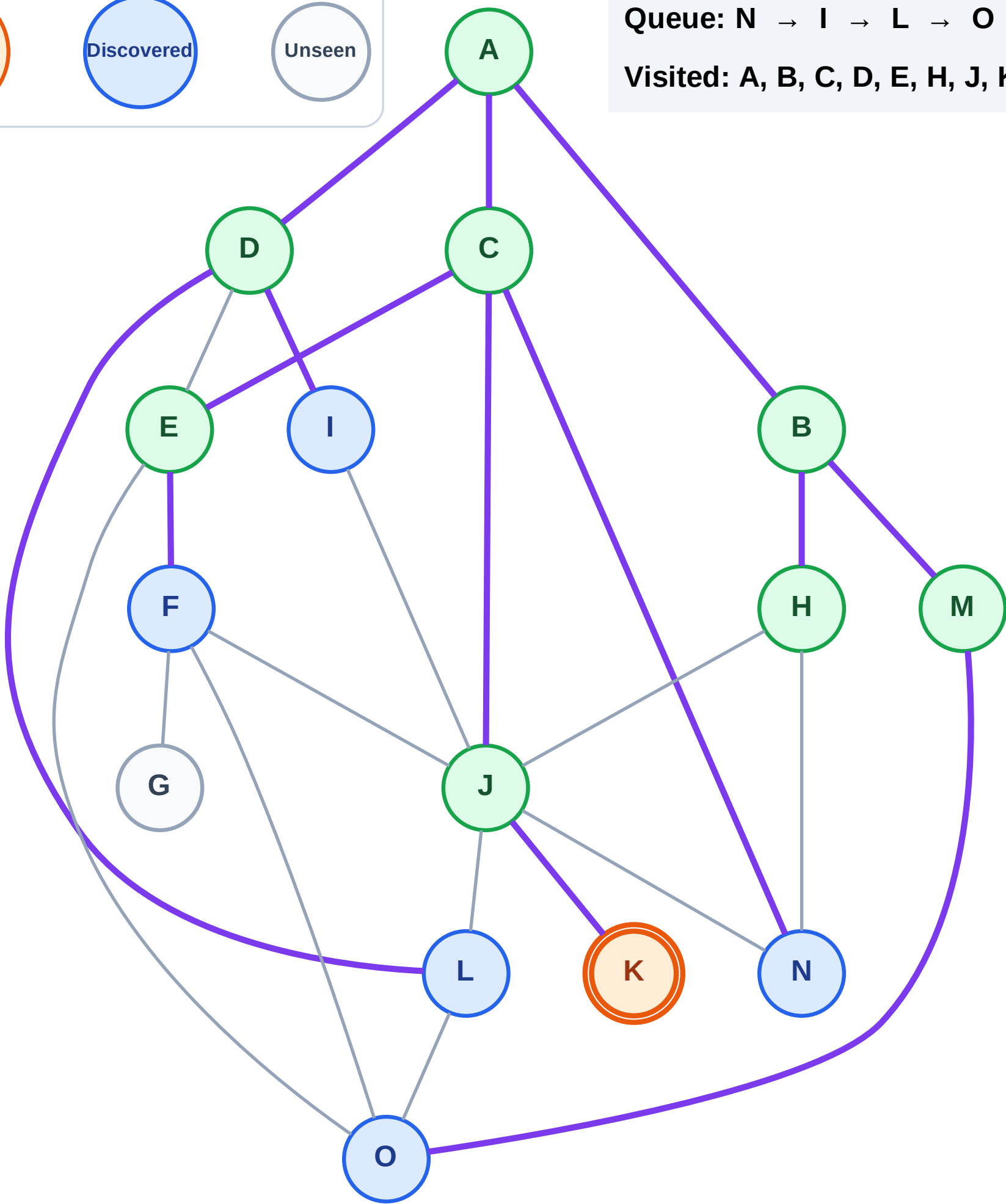
Step 17: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → C → J → K

Legend



Queue: N → I → L → O → F

Visited: A, B, C, D, E, H, J, K, M



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

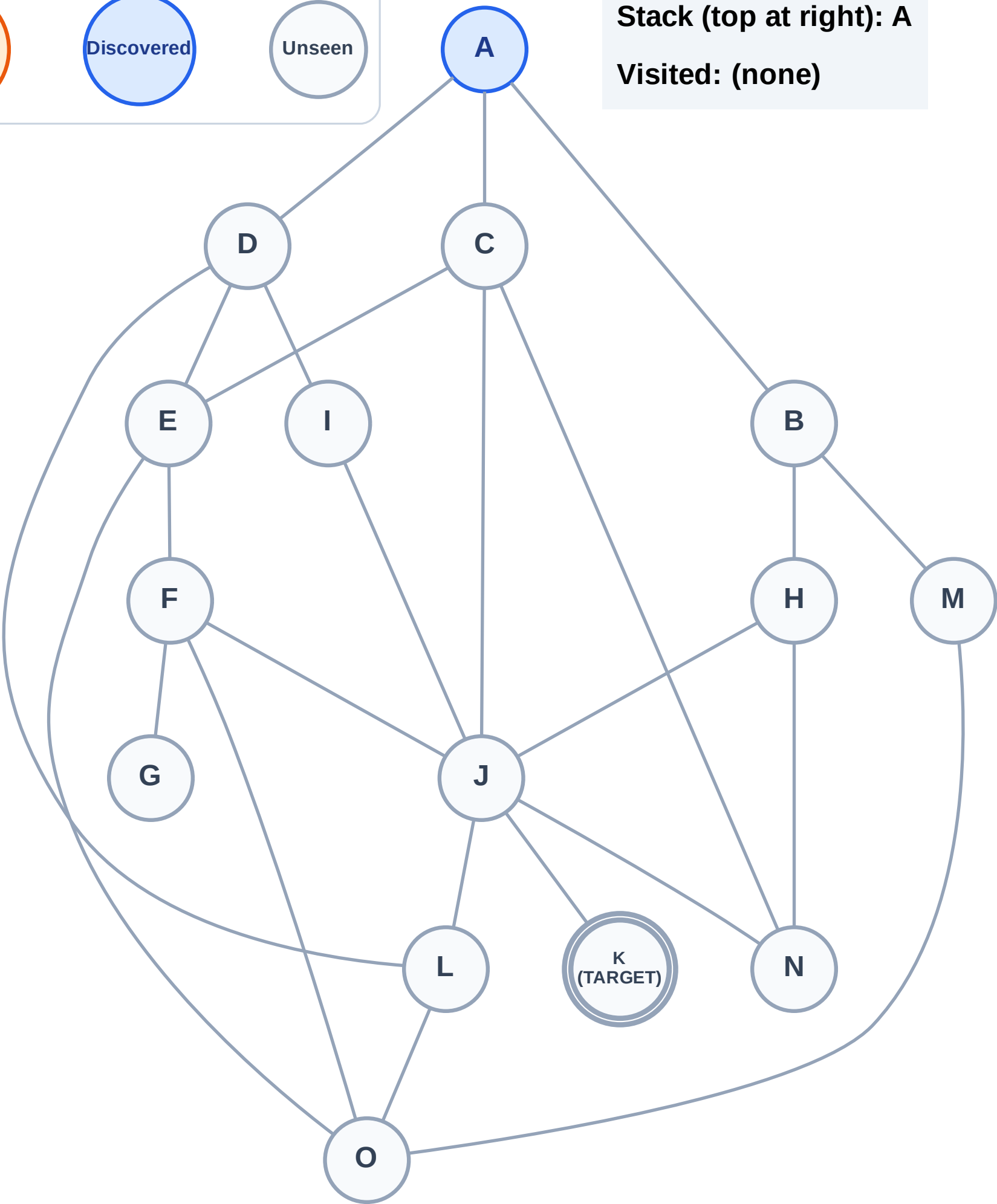
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

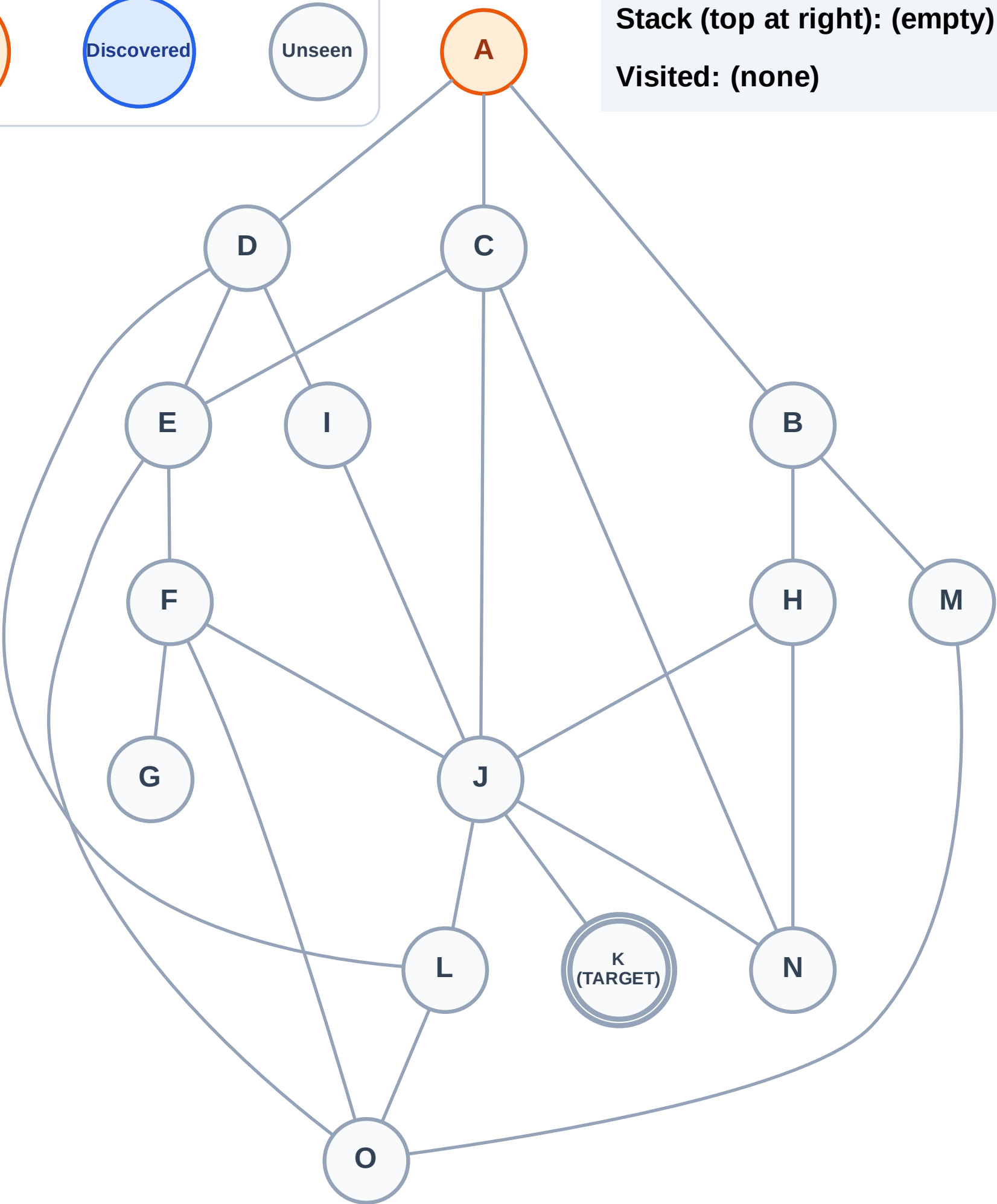
Processing

Discovered

Unseen

Stack (top at right): (empty)

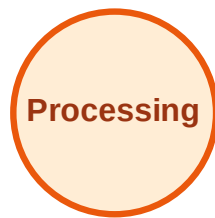
Visited: (none)



DFS Traversal

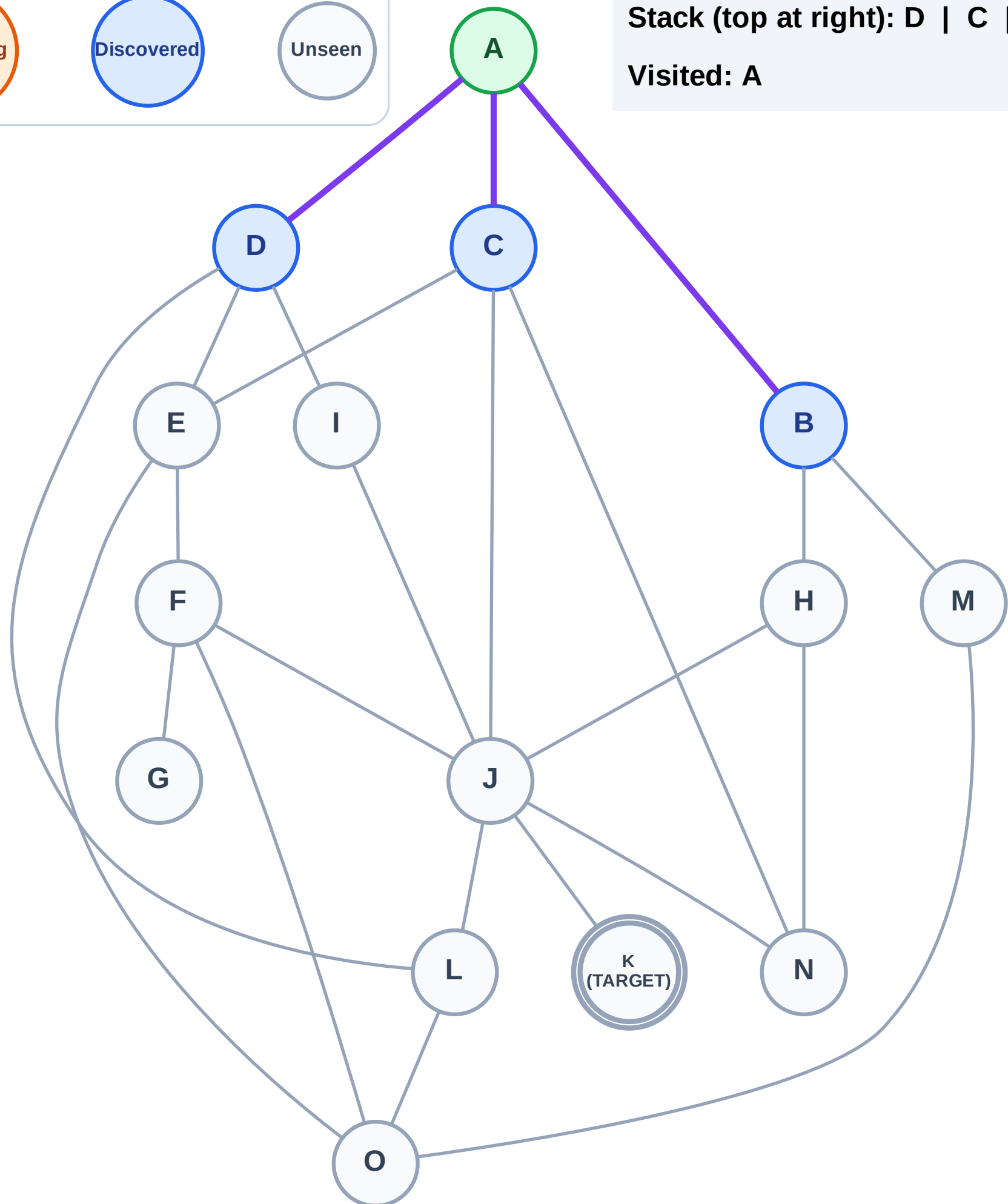
Step 3: Discover D, C, B from A

Legend



Stack (top at right): D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

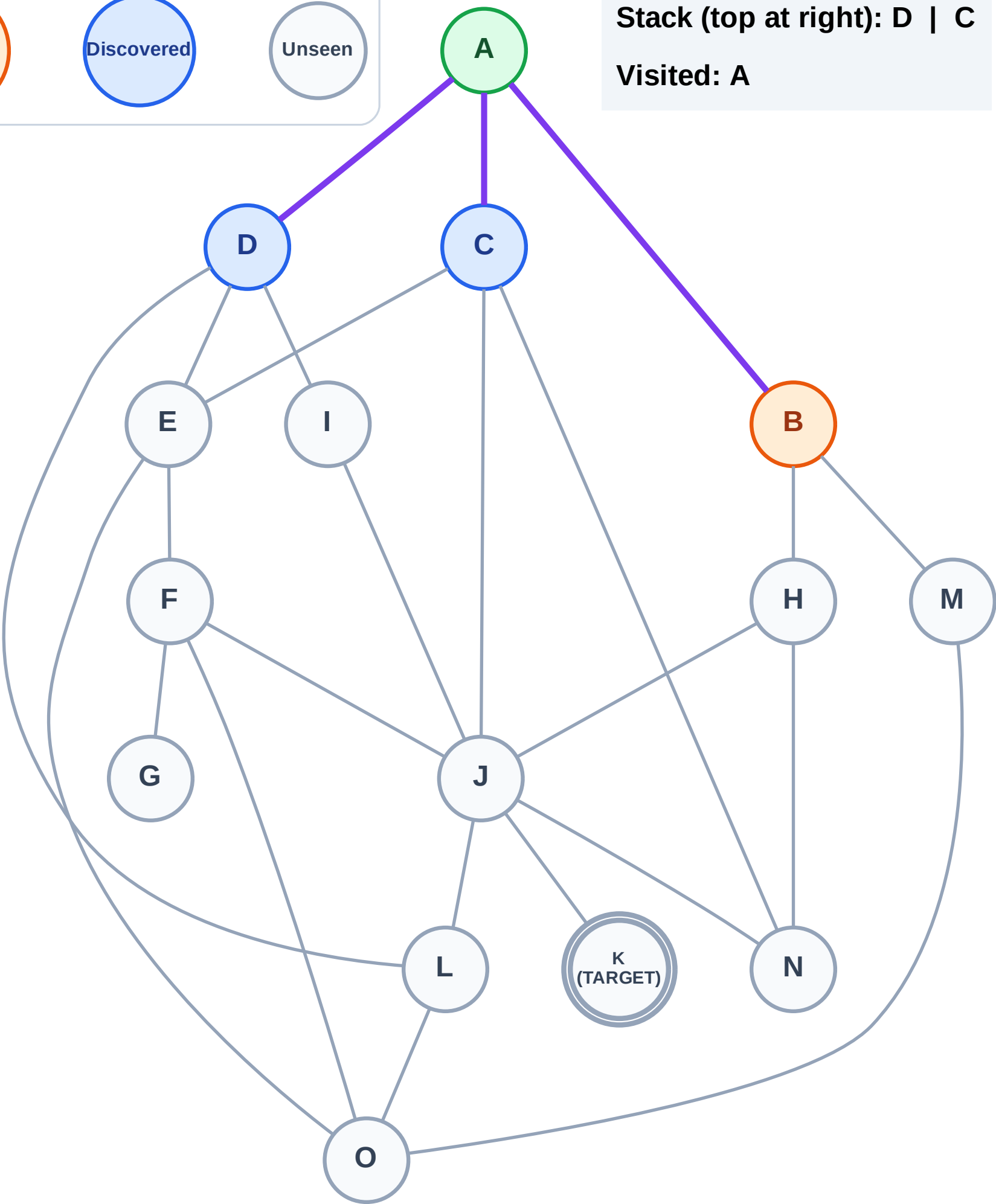
Processing

Discovered

Unseen

Stack (top at right): D | C

Visited: A



DFS Traversal

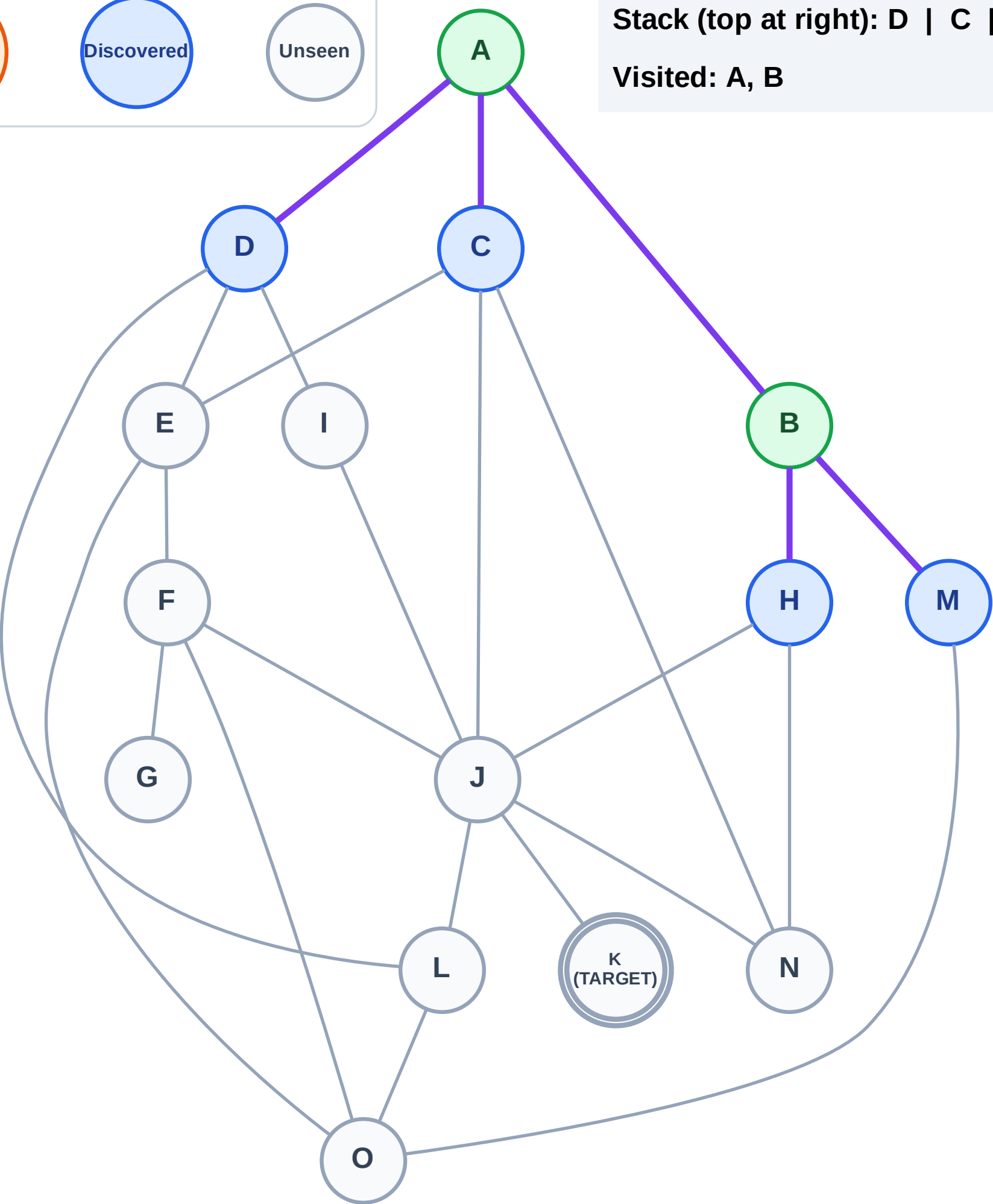
Step 5: Discover M, H from B

Legend

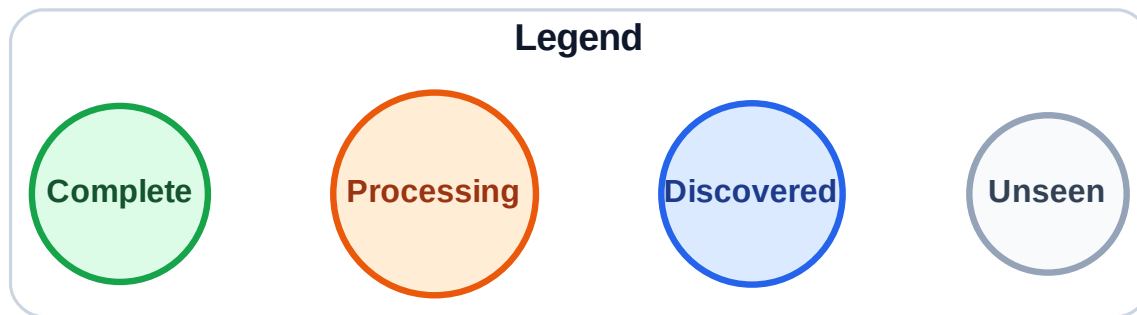


Stack (top at right): D | C | M | H

Visited: A, B

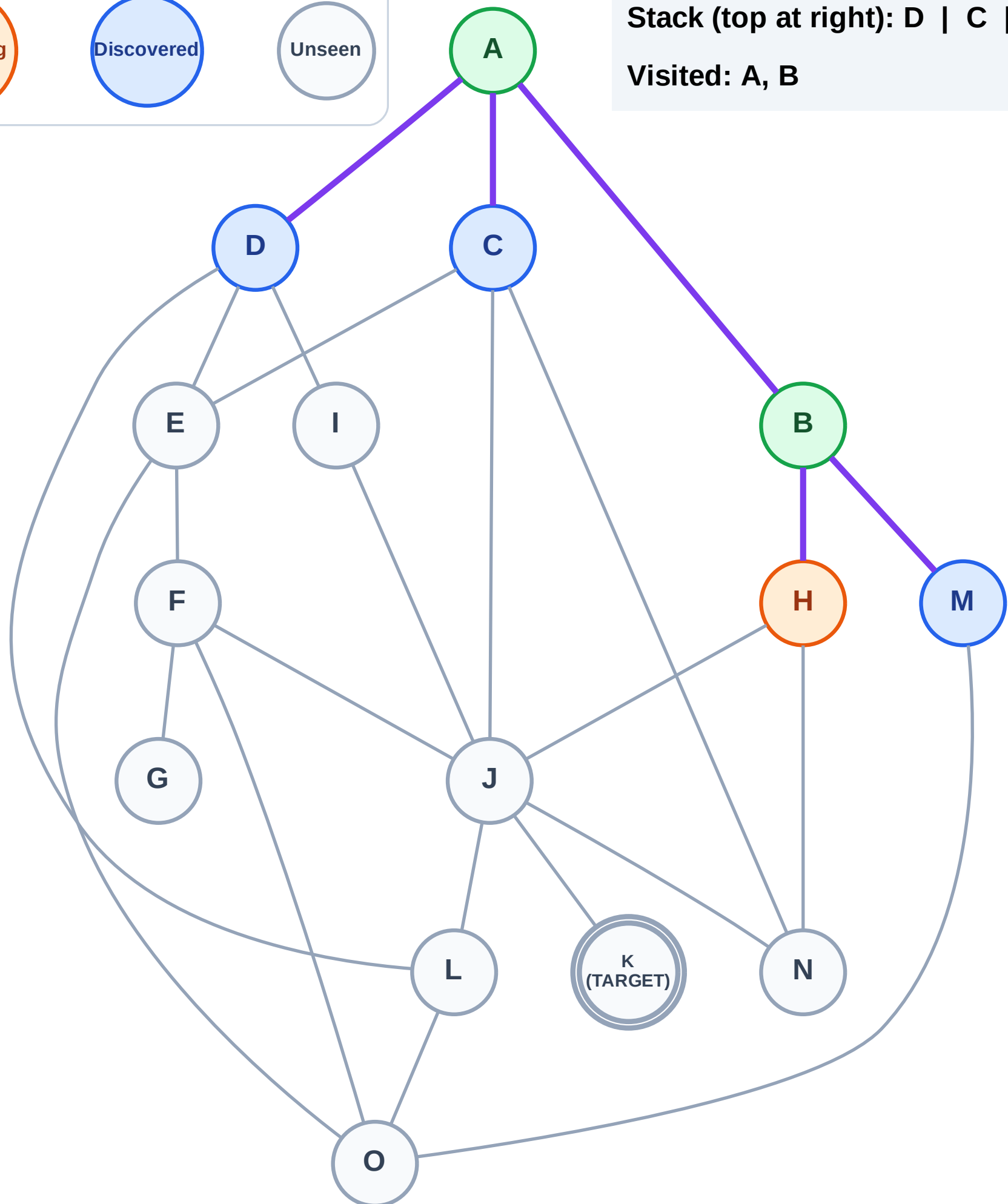


Step 6: Process node H



Stack (top at right): D | C | M

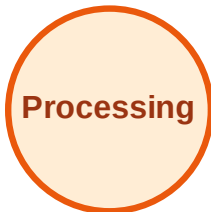
Visited: A, B



DFS Traversal

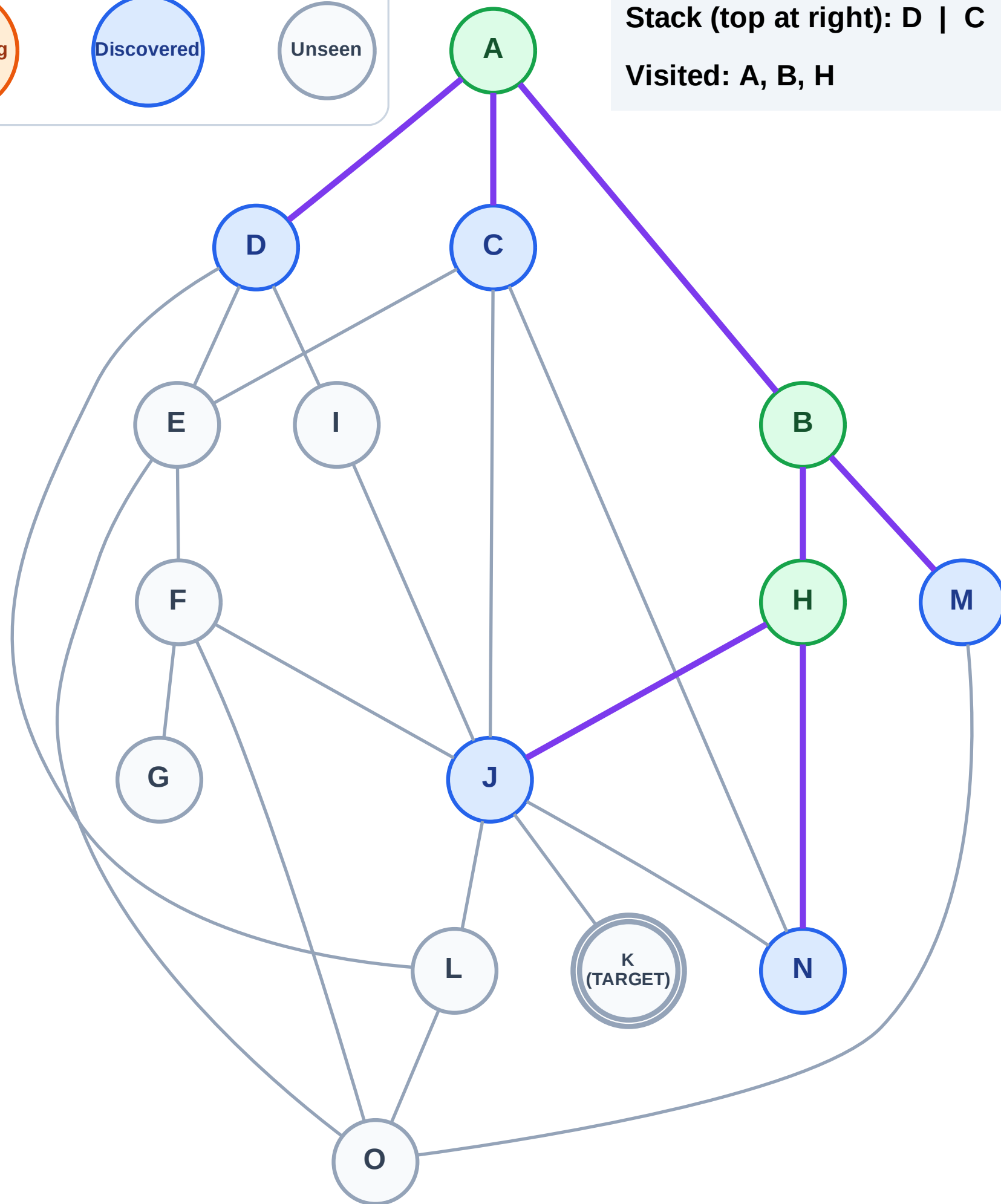
Step 7: Discover N, J from H

Legend



Stack (top at right): D | C | M | N | J

Visited: A, B, H



DFS Traversal

Step 8: Process node J

Legend

Complete

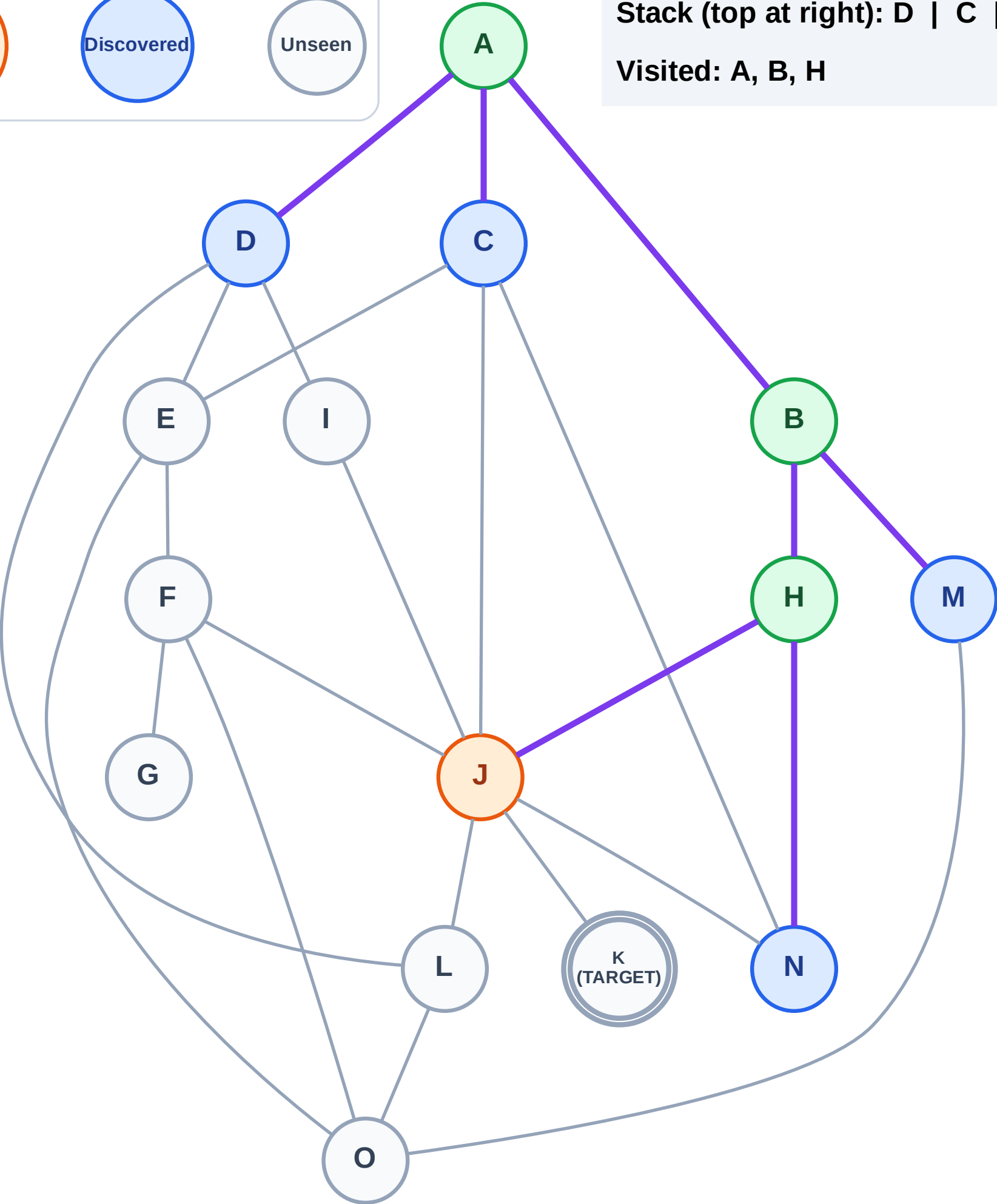
Processing

Discovered

Unseen

Stack (top at right): D | C | M | N

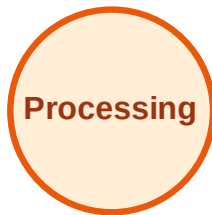
Visited: A, B, H



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → H → J → K

Legend



Stack (top at right): D | C | M | N | L

Visited: A, B, H, J, K

